Nonparametric Intensity Estimation in Covariate-Driven Poisson Processes Using Deep Learning

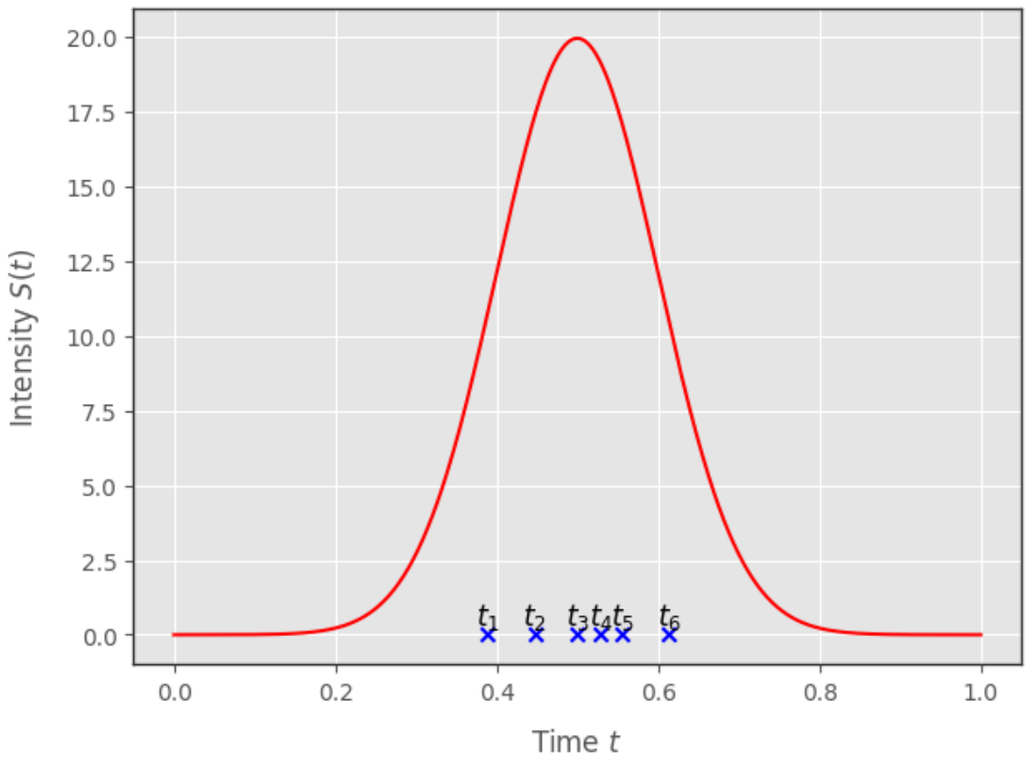
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Overview



Problem: The excess risk of MLE (KL div.) diverges easily
→ Makes theoretical analysis difficult

Solution: Replace KL divergence with Hellinger distance

Derive an inequality that evaluates the performance of the NPMLE under weaker assumptions (Theorem 1)

Covariate-driven Poisson process

$Z \sim P_Z,\ X \mid Z \sim \text{Poisson process}(S(\cdot \mid Z))$

Observations: $\{(Z_i, X_i)\}_{i=1}^n$

Estimating the intensity function $S(\cdot \mid \cdot)$

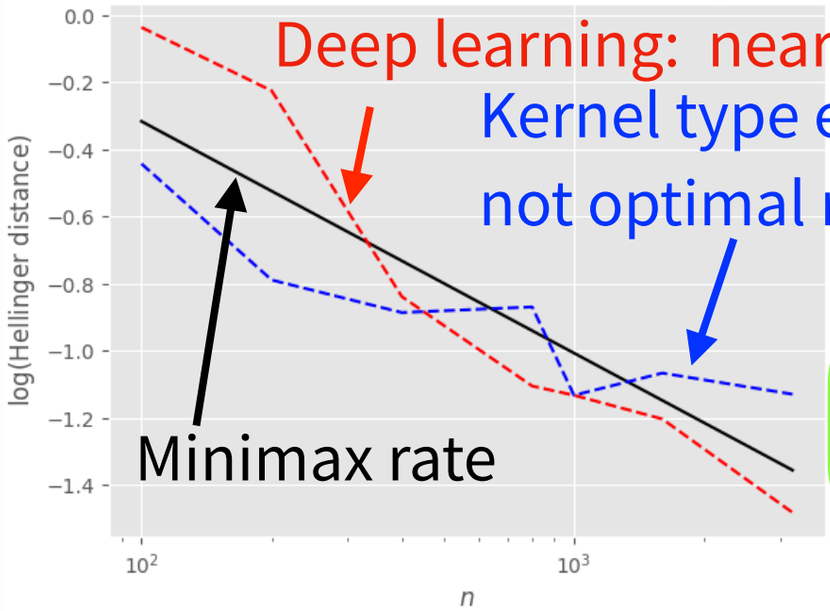
is crucial in applications

e.g., ambulance demand forecasting

→ optimal placement of ambulances

Apply Theorem 1 to show convergence rates for deep-learning–based MLE

Derive the minimax optimality (Theorem 2)



Log-scale plot

Poisson processes

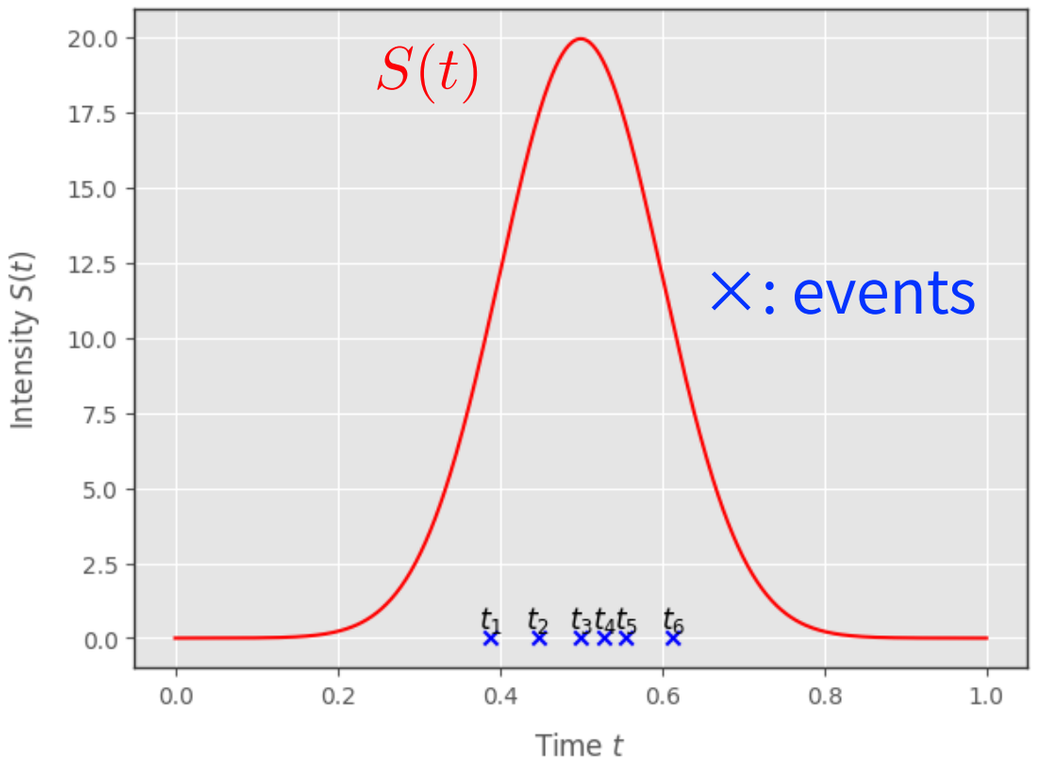
Poisson Processes and Intensity Functions

- A Poisson process models random events occurring in time or space
- Definition of Poisson process X (on $[0, 1]$, for simplicity)
  - X is a non-negative integer-valued random measure $[0, 1]$
  - $X(\{0\}) = 0,\ a.s.$
  - For $0 \le t_1 < \cdots < t_N \le 1$, $X([t_1, t_2)), \ldots, X([t_{N-1}, t_N))$ are independent random variables.
  - For any interval $[s, t)$,

$$X([s, t\}) \sim \text{Poisson}(\int_s^t S(\tau)\, d\tau)$$

intensity function

- The intensity function describes how likely events are to occur at time t
- Once the intensity is known, the Poisson process is determined



Poisson Processes with Covariates

⭐Conditional intensity functions

- We consider intensity functions that vary depending on Z

$$X \mid Z \sim \text{Poisson process}(S(\cdot \mid Z))$$

Conditional intensity function

- This is a special case of Cox processes,
  where the intensity itself is random
- In real data, intensities naturally change due to many factors

Application	Intensity meaning	Covariates
Wildfire ignition locations (D'Angelo et al., 2025)	Likelihood of ignition at a coordinate	Temperature, humidity, wind speed
Crime locations (Hart et al., 2022)	Crime incidence rate at a coordinate	Temperature, precipitation

Maximum Likelihood Estimation for Conditional Intensities

⭐ Nonparametric maximum likelihood estimator (NPMLE)

- $A \subset \mathbb{R}^{d_1}$: space of events, $C \subset \mathbb{R}^{d_2}$: space of covariates, $d = d_1 + d_2$
- $(Z_1, X_1), \ldots, (Z_n, X_n)$: i.i.d. observations of covariates and events

$$Z_i \sim_{\text{i.i.d.}} P_Z, \quad X_i \mid Z_i \sim \text{Poisson process}(S_0(\cdot \mid Z_i))$$

- S_0: underlying (true) intensity function
- Negative log likelihood function:

$\boldsymbol{x}_{ij}$: events coresspondingto X_i
$N_i = X_i(A)$, ν: measure on A

$$\ell_n(X_1, \ldots, X_n, \boldsymbol{z}_1, \ldots, \boldsymbol{z}_n \mid S) \coloneqq \frac{1}{n} \sum_{i=1}^{n} \left\{ -\sum_{j=1}^{N_i} \log(S(\boldsymbol{x}_{ij} \mid \boldsymbol{z}_i)) + \int_A S(\boldsymbol{x} \mid \boldsymbol{z}_i)\, d\nu(\boldsymbol{x}) \right\}$$

- $\mathcal{F}_n$: hypothesis space
  - e.g., DNN, kernel-type estimator (Klutchnikoff & Massiot, 2016)
- Nonparametric maximum likelihood estimator (NPMLE):

$$\hat{S}_n \in \min_{S \in \mathcal{F}_n} \ell_n(S)$$

- The integral can be approximated via Monte Carlo

Difficulties in Nonparametric Maximum Likelihood Estimation

⭐ Excess risk analysis of MLE in various settings

- **Regression**
  - If errors are Gaussian, the excess risk reduces to the squared error
  - If the true regression function is bounded, the squared error is finite
  - **This boundedness assumption is reasonable**
- **Classification**
  - The excess risk involves the **KL divergence** between the true and estimated conditional probabilities
    - **KL divergence is generally unbounded and can diverge easily**
  - **To evaluate the excess risk, we need additional assumptions**
    - e.g., the true conditional probability is bounded away from zero

$$c < \mathbb{P}(Y = 1 \mid \boldsymbol{x}) < C, \quad 0 < c, C < 1$$

- **Estimating intensity functions**
  - **Just like classification, KL divergence diverges easily**

Difficulties in Nonparametric Maximum Likelihood Estimation

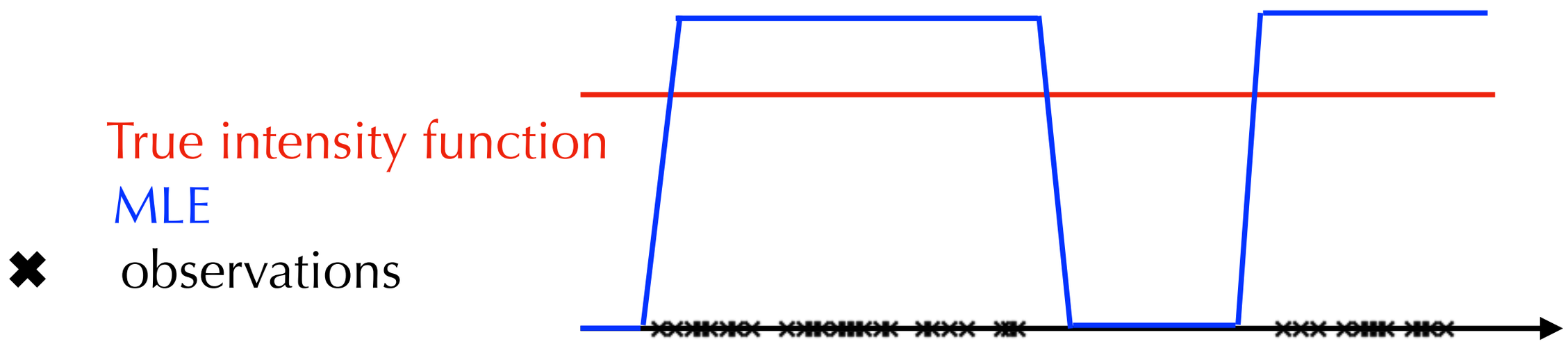
⭐Example where the KL divergence diverges

- For Poisson processes, the KL divergence involves **the ratio of intensity functions**:

$$\mathrm{KL}(P_f, P_g) = \int_A \left(f \log\left(\frac{f}{g}\right) + g - f \right) d\nu$$

P_h: the distribution of a Poisson process with intensity function h

- If the support of the estimator does not cover the support of the true intensity function, the KL divergence becomes infinite



Related Works

⭐Related works

- Concerning parametric estimation of the intensity function of the Poisson process, good properties such as consistency and asymptotic normality hold. (Kutoyants, 1998)
- Nonparametric intensity estimation theory remains limited

	Estimator	Model	Convergence rates	Underlying function class	Notes
Flexman et al. (2017)	MLE	RKHS	-	-	-
Kirichenko & Zanten (2015)	Bayesian	Hierarchical modeling	$n^{-\frac{2\beta}{2\beta+d}}$	Hölder	Bounded away from zero Minimax optimal
This work	**MLE**	**Deep learning**	$n^{-\frac{\beta}{\beta+d}}$	Composition of Hölder	**Minimax optimal**

Goals of This Study

⭐To clarify the theoretical properties of the NPMLE of intensity functions for Poisson processes with covariates

- Issue: KL divergence diverges easily → standard analysis fails
- Approach: Evaluate Hellinger distance instead

⭐Analyze deep-learning–based NPMLE

- If the true intensity is composite structured
  → avoids curse of dimensionality, achieves nearly optimal rate
- If supported on a Riemannian manifold
  → automatically adapt to low-dimensional
  Riemannian manifold structures

Notations & Assumptions

⭐Notations

- $\tilde{H}^2(S_1, S_2) := \mathbb{E}_Z \left[\frac{1}{2} \int_A (\sqrt{S_1(\boldsymbol{x} \mid Z)} - \sqrt{S_2(\boldsymbol{x} \mid Z)})^2 d\boldsymbol{x} \right]$
  - **Instead of KL divergence between Poisson processes, we evaluate $\tilde{H}^2$**
- $\mathcal{F}(L, r, B)$: class of DNN
  - L : depth, r : width per layer, B : parameter magnitude bound
  - Activations: ReLU everywhere, including the output layer

⭐Assumption

$$\exists c_0 > 0,\ \forall n \in \mathbb{N},\ \exists S_n^* \in \mathcal{F}_n,\ \forall \boldsymbol{x} \in A,\ \forall \boldsymbol{z} \in C;\ \frac{S_0(\boldsymbol{x} \mid \boldsymbol{z})}{S_n^*(\boldsymbol{x} \mid \boldsymbol{z})} \leq c_0^2$$

- This holds automatically when using DNN estimator

Oracle Inequality

★Oracle inequality

Theorem1 (Oracle Inequality)

We consider the estimation of the conditional intensity function of a Poisson process. Suppose that δ_n and a universal constant c satisfies the following inequality:

$$\sqrt{n}\delta_n^2 \geq c\Psi(\delta_n)$$

Then, for any $\delta \geq \delta_n$, the following inequality holds.

$$\mathbb{E}[\tilde{H}^2(\hat{S}_n, S_0)] \leq 1224(c_0+1)^2 \left(\underbrace{\delta^2}_{\text{Variance}} + \frac{c^3}{n} + \underbrace{\tilde{H}^2(S_n^*, S_0)}_{\text{Bias}^2} \right)$$

- Ψ: model complexity (e.g., covering number)
- The NPMLE convergence rate depends on
  the Complexity (Variance) term + the Approximation (Bias) term

Structure of the True Intensity Function

⭐Function classes of the true intensity functions

- Hölder class with smoothness β, domain D, and bound Q: $\mathcal{C}^{\beta}(D, Q)$
- Composite function class:

$$\mathcal{G}_{\text{comp}}(q, \mathbf{d}, \mathbf{t}, \boldsymbol{\beta}, Q) = \{f = \boldsymbol{g}_q \circ \cdots \circ \boldsymbol{g}_0 : \boldsymbol{g}_i = (g_{ij})_j : [a_i, b_i]^{d_i} \to [a_{i+1}, b_{i+1}]^{d_{i+1}},$$
$$g_{ij} \in \mathcal{C}^{\beta_i}([a_i, b_i]^{t_i}, Q), \text{for some} |a_i|, |b_i| \leq Q\}$$

  - $f(x_1, x_2, x_3) = g_1 \circ \boldsymbol{g}_2(x_1, x_2, x_3) = g_{11}(g_{01}(x_1, x_3), g_{02}(x_1, x_2))$
    3 dim. 2 dim.
  - **There is a low-dimensional structure hidden inside**
- Intensity supported on a low-dimensional manifold
  - $D = \mathcal{M} \times C$
  - $\mathcal{M} \subset A$ is a compact Riemannian manifold with $d_{\mathcal{M}} < d_1$
    - $\dim(\mathcal{M}) = d_{\mathcal{M}}$

Convergence Rates for Deep Learning Estimators

⭐Convergence rates for DNN-based MLE

- Using Theorem 1, we obtain convergence rates for fixed function classes of the true intensity function

function class	$\mathcal{C}^{\beta}(\mathcal{M} \times C, Q)$	$\mathcal{G}_{\mathrm{comp}}$
convergence rates (exponent of n)	$-\dfrac{\beta}{\beta + d_{\mathcal{M}} + d_2}$	$-\max\limits_{i=0,\ldots,q} \dfrac{\beta_i^*}{\beta_i^* + t_i}$

β_i^* : smoothness

- **Notably, the convergence rates do not depend on the ambient dimension d**
- **DNNs capture intrinsic low-dimensional structure**
  → Avoid curse of dimensionality

Minimax Optimality

⭐Minimax lower bound

Theorem 2 (Minimax lower bound)

Z_j are drawn from a distribution with Lebesgue density on $[0,1]^{d_2}$, which is lower and upper bounded by positive constants. For any nonnegative integer q, any $\boldsymbol{\beta} \in \mathbb{R}^r$, any $\mathbf{d}, \mathbf{t}$ satisfying $t_i \leq \min(d_0, \ldots, d_{i-1})$ for all i, and a sufficiently large constant $Q > 0$, there exists a positive constant c such that

$$\inf_{\hat{f}_n} \sup_{f_0 \in \mathcal{G}_{\text{comp}}(q,\mathbf{d},\mathbf{t},\boldsymbol{\beta},Q)} \tilde{H}^2(\hat{S}_n, S_0) \geq c\phi_n,$$

where the infimum is taken over all estimators.

- The convergence rate derived in the previous slide matches this (up to log-factor) → **Our rate is minimax optimal**

Sparsity in DNN

⭐Sparsity in DNN

- Many papers assume neural network sparsity,
  i.e., few nonzero parameters
  - 📌 e.g., Suzuki (2019), Schmidt-Hieber (2020)
- Why? ⇒ excess risks are typically bounded by:

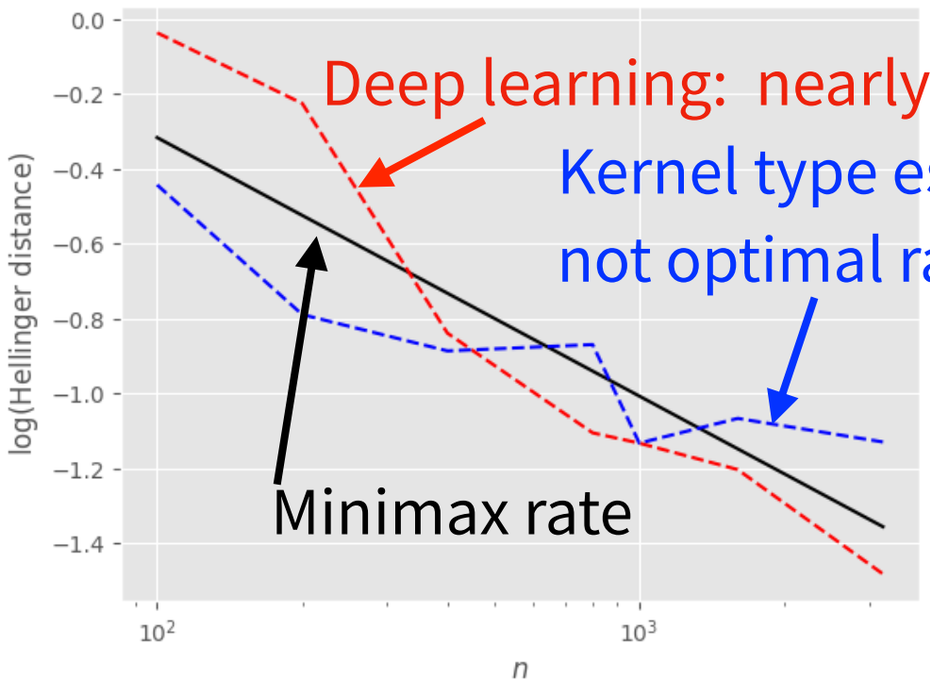
$$\frac{\text{“Complexity”}}{n} + \text{Bias}^2$$

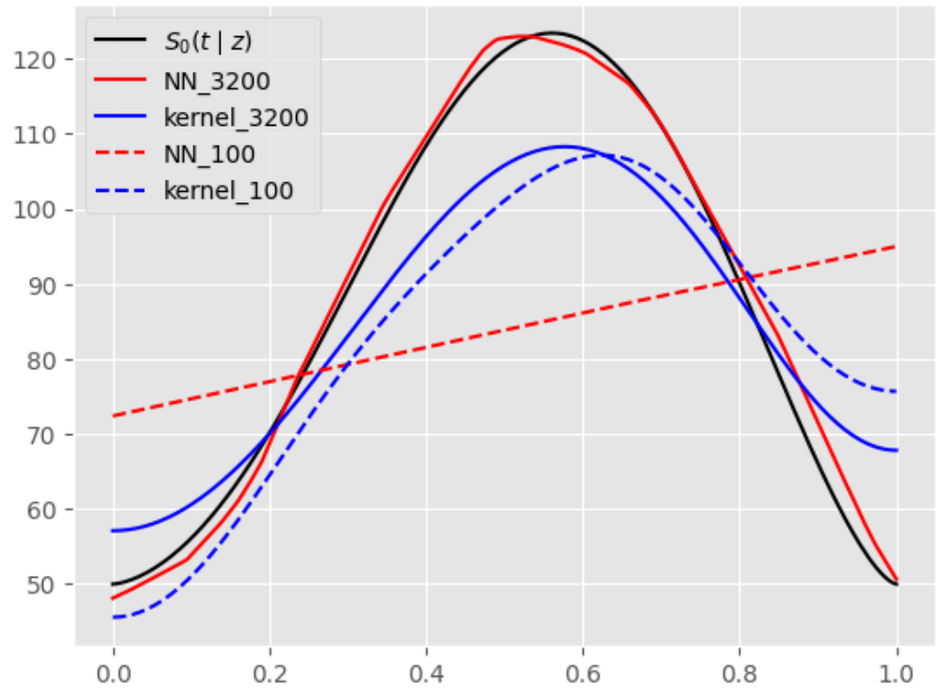
- “Complexity”: covering number, bracketing number
- Covering number scales with the number of parameters
  → Sparsity leads to a tighter bound
- **Using tighter bias control (Kohler & Langer, 2021), we achieve fast rates without requiring sparsity**

Numerical Experimtents

⭐When the true intensity function is in $\mathcal{G}_{\mathbf{comp}}$

- $A = [0,1]$, $C = [0,1]^4$, $d = 5$
- $\boldsymbol{Z} = (Z_1, Z_2, Z_3, Z_4)$, $Z_1, Z_2, Z_3, Z_4 \sim_{\text{i.i.d.}} \text{Unif}(0,1)$
- $S_0(x \mid \boldsymbol{Z}) = 10(5 + Z_1 f(x - Z_2) + Z_3 f(x - Z_4))$
  - $f \in \mathcal{C}^{1.5}([0,1], Q)$
- The convergence rate from Theorem 1 is $n^{-3/10}$
  - **Note that this rate is the minimax optimal rate**
- Comparison with kernel-type estimator





Summary

- ★**We establish theoretical guarantees for NPMLE of covariate-driven Poisson intensities (Theorem 1)**

- ★**We derive convergence rates for deep learning-based MLE**
  - Deep learning can avoid the curse of dimensionality
  - Minimax optimality is proved

function class	$\mathcal{C}^{\beta}(\mathcal{M} \times C, Q)$	$\mathcal{G}_{\mathrm{comp}}$
convergence rates (exponent of n)	$-\dfrac{\beta}{\beta + d_{\mathcal{M}} + d_2}$	$-\max\limits_{i=0,\ldots,q} \dfrac{\beta_i^*}{\beta_i^* + t_i}$

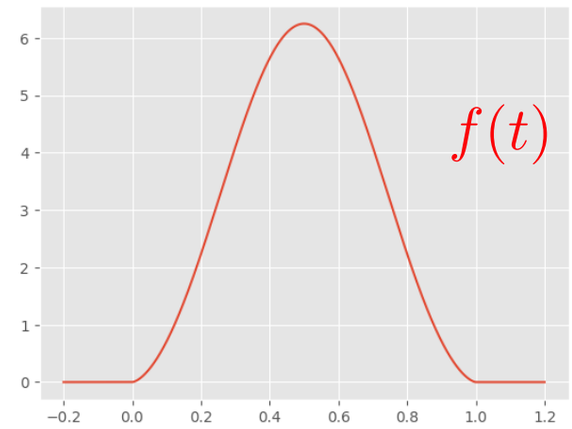
Minimax optimal

Appendix

Experiment Setup Details

⭐When the true intensity function is in $\mathcal{G}_{\text{comp}}$

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- $S_0(x \mid \boldsymbol{Z}) = 10(5 + Z_1 f(x - Z_2) + Z_3 f(x - Z_4))$
  - $f \in \mathcal{C}^{1.5}([0,1], Q)$



- DNN: AdamW + early stopping
- Kernel: 5-fold cross-validation
- For each sample size, both models trained 3 times, median reported
- Training time for n = 1000
  - Kernel: 700 sec，DNN: 109 sec

Convergence rates

Theorem (Convergence rates)

(1) For $\mathcal{G}_{\text{comp}}$, if we take

$$\text{(i)}\quad L \asymp \log(n), \quad \text{(ii)}\quad \sqrt{n\phi_n} \asymp r, \quad \text{(iii)}\ B \asymp \max_{i=0,\ldots,q} (n\phi_n)^{\frac{2(\beta_i+1)}{t_i}},$$

$$\mathbb{E}[\widetilde{H}^2(\hat{S}_n, S_0)] \leq C\phi_n \log(n)^3.$$

where $\beta_i^* = \beta_i \prod_{l=i+1}^{r} (\beta_l \wedge 1)$, $\phi_n = \max_{i=0,\ldots,r} n^{-\frac{\beta_i^*}{\beta_i^*+t_i}}$.

(2) For $\mathcal{C}^{\beta}(\mathcal{M} \times C, Q)$, if we take

$$\text{(i)}\quad L \asymp \log(n), \quad \text{(ii)}\quad \sqrt{n\phi'_n} \asymp r, \quad \text{(iii)}\ B \asymp (n\phi'_n)^{\frac{2(\beta+1)}{d_{\mathcal{M}}+d_2}},$$

$$\mathbb{E}[\widetilde{H}^2(\hat{S}_n, S_0)] \leq C'\phi'_n \log(n)^3.$$

where $\phi'_n = n^{-\frac{(1+\alpha)\beta}{(1+\alpha)+d_{\mathcal{M}}+d_2}}$.